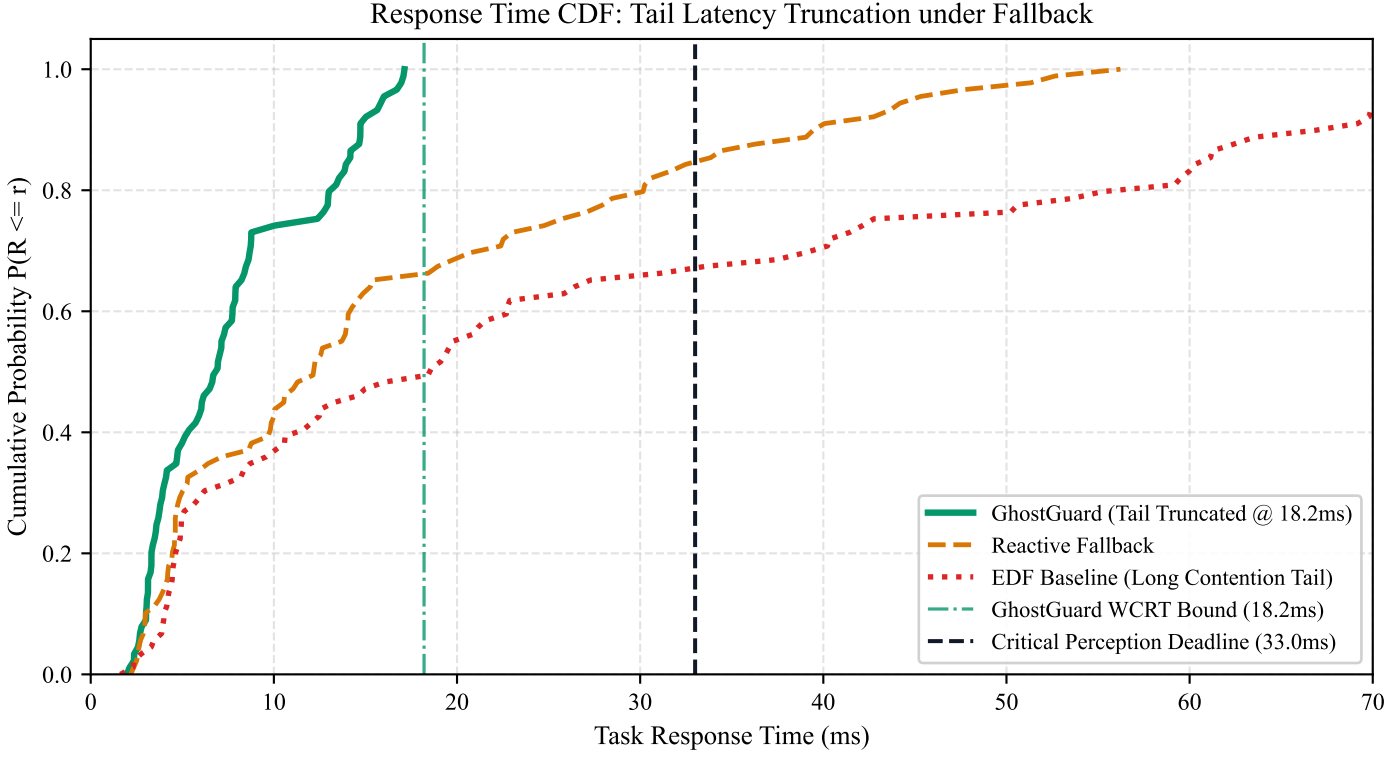
Response Time CDF: Tail Latency Truncation under Fallback
Cumulative Probability P(R <= r)
Task Response Time (ms)
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
60
70
GhostGuard (Tail Truncated @ 18.2ms)
Reactive Fallback
EDF Baseline (Long Contention Tail)
GhostGuard WCRT Bound (18.2ms)
Critical Perception Deadline (33.0ms)